

Screen-based entertainment time, all-cause mortality, and cardiovascular events: population-based study with ongoing mortality and hospital events ...

OBJECTIVES: The aim of this study was to examine the independent relationships of television viewing or other screen-based entertainment ("screen time") with all-cause mortality and clinically confirmed cardiovascular disease (CVD) events. A secondary objective was to examine the extent to which metabolic (body mass index, high-density lipoprotein and total cholesterol) and inflammatory (C-reactive protein) markers mediate the relationship between screen time and CVD events.

BACKGROUND: Although some evidence suggests that prolonged sitting is linked to CVD risk factor development regardless of physical activity participation, studies with hard outcomes are scarce.

METHODS: A population sample of 4,512 (1,945 men) Scottish Health Survey 2003 respondents (≥ 35 years) were followed up to 2007 for all-cause mortality and CVD events (fatal and nonfatal combined). Main exposures were interviewer-assessed screen time (< 2 h/day; 2 to < 4 h/day; and ≥ 4 h/day) and moderate to vigorous intensity physical activity. **RESULTS:** Two hundred fifteen CVD events and 325 any-cause deaths occurred during 19,364 follow-up person-years. The covariable (age, sex, ethnicity, obesity, smoking, social class, long-standing illness, marital status, diabetes, hypertension)-adjusted hazard ratio (HR) for all-cause mortality was 1.52 (95% confidence interval [CI]: 1.06 to 2.16) and for CVD events was 2.30 (95% CI: 1.33 to 3.96) for participants engaging in ≥ 4 h/day of screen time relative to < 2 h/day. Adjusting for physical activity attenuated these associations only slightly (all-cause mortality: HR: 1.48, 95% CI: 1.04 to 2.13; CVD events: HR: 2.25, 95% CI: 1.30 to 3.89). Exclusion of participants with CVD events in the first 2 years of follow-up and previous cancer registrations did not change these results appreciably. Approximately 25% of the association between screen time and CVD events was explained collectively by C-reactive protein, body mass index, and high-density lipoprotein cholesterol. **CONCLUSIONS:** Recreational sitting, as reflected by television/screen viewing time, is related to raised mortality and CVD risk regardless of physical activity participation.

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